The Arrow of Time as Petz Recovery Failure

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The Petz recovery map plays a central role across quantum information, error correction, and thermodynamics—a fact well recognized by these communities. Here we offer a physical interpretation of this mathematical unity: we define a temporal asymmetry parameter $\tau = 1 - F$, where F is the fidelity of Petz recovery, and show that τ continuously quantifies the arrow of time. For closed systems $\tau = 0$: forward and reverse processes are indistinguishable. For open systems $\tau > 0$, bounded by entropy production via $\tau \leq 1 - \exp(-\Sigma/2)$ (a consequence of the universal recovery bound of Junge et al.). This interpretation unifies the quantum eraser ($\tau = 0$ because the signal-idler pair is closed), quantum error correction ($\tau > 0$ sets the decoder's task), and thermodynamic irreversibility (τ bounded by dissipation) under a single physical picture. We present two new results. First, we identify sufficient conditions under which the recovery bound is exactly saturated ($F^2 = \exp(-\Delta D)$): the saturation value is a Bayesian posterior ratio, connecting quantum recovery to classical retrodiction. Second, we prove that $\sqrt{\tau}$ satisfies a triangle inequality under channel composition—a property unique to the Petz map that fails for the universal (rotated Petz) recovery map of Junge et al.

INTRODUCTION

The Petz recovery map [3, 4] plays a central role across several areas of quantum information science: as a near-optimal quantum error correction (QEC) decoder [5]; as the *unique* retrodiction functor in categorical quantum mechanics [6]; as the reverse channel in quantum fluctuation theorems [7]; as a physical reversal via dilations [17]; and in entanglement wedge reconstruction in holography [8]. These connections are well established in the literature, and recent work by Buscemi, Fullwood, and Parzygnat [31] has further developed the information-theoretic foundations linking them.

In this Letter, we offer a physical interpretation of this mathematical unity. We define a temporal asymmetry parameter $\tau = 1 - F$, where F is the Petz recovery fidelity, and argue that τ provides a continuous, operational measure of the arrow of time.

The interpretive framework rests on known results assembled into a single picture: (1) the quantum eraser [1, 2] is a physical instantiation of Petz recovery; (2) the uniqueness of the Petz map as retrodiction functor (due to Parzygnat–Buscemi [6]) explains why it also serves as a near-optimal QEC decoder [5, 11]; (3) the universal recovery bound of Junge et al. [11] (Corollary 5.1 therein) and the Fawzi–Renner inequality [12], combined with the Kolchinsky–Wolpert decomposition [13] and the Reeb–Wolf bound [18], yield a chain linking recovery fidelity to entropy production. The logical flow is:

$$\text{Eraser} \xrightarrow{\text{Petz}} \text{Retrodiction} \to \text{QEC} \to \text{Thermo}. \qquad (1)$$

The physical content is that the arrow of time—the empirical asymmetry between past and future—can be identified with Petz recovery failure. For a closed system, the Petz map recovers the initial state exactly ($F = 1$, $\tau = 0$); there is no distinction between forward and reverse processes. Openness to an environment ($\Sigma > 0$) breaks this symmetry, and $\tau \leq 1 - \exp(-\Sigma/2)$ quantifies the cost.

We present two new mathematical results. First, we identify sufficient conditions under which the Junge et al. bound $F^2 \geq \exp(-\Delta D)$ is exactly saturated, with a saturation value given by a Bayesian likelihood ratio (Theorem 4). Second, we prove that $\sqrt{\tau}$ satisfies a triangle inequality under channel composition—a structural property unique to the Petz map that fails for the rotated Petz map (Theorem 5).

THE PETZ RECOVERY MAP

Given a quantum channel $\mathcal{N} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ and a faithful reference state $\sigma \in \mathcal{S}(\mathcal{H}_A)$, the *Petz recovery map* is

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) = \sigma^{1/2}\, \mathcal{N}^\dagger\big(\mathcal{N}(\sigma)^{-1/2}\, \rho\, \mathcal{N}(\sigma)^{-1/2}\big)\, \sigma^{1/2}, \qquad (2)$$

where $\mathcal{N}^\dagger$ is the Hilbert–Schmidt adjoint of $\mathcal{N}$. The map $\mathcal{R}_{\sigma,\mathcal{N}}$ is completely positive and trace-preserving (CPTP) [3], and satisfies exact recovery $\mathcal{R}_{\sigma,\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on the support of σ if and only if the data-processing inequality (DPI) for relative entropy is saturated: $D(\sigma\|\tau) = D(\mathcal{N}(\sigma)\|\mathcal{N}(\tau))$ for all τ [4].

Table I shows how the mathematical ingredients of Eq. (2) map to physical meaning in each domain.

UNIQUENESS OF RETRODICTION

We recall the uniqueness result of Parzygnat and Buscemi [6], which is central to our interpretation.

TABLE I. Physical interpretation of the Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ across four domains.

	Eraser	**Time Rev.**	**QEC**	**Thermo.**
$\mathcal{N}$	Path mark	Fwd. evol.	Noise	Irrev. proc.
σ	Entangled	Prior	Code state	Gibbs
$\mathcal{R}$	Erasure	Retrodiction	Decoder	Rev. proc.
ΔD	Decoherence	Info. loss	Code dist.	βW_{diss}

Theorem 1 (Retrodiction Uniqueness (Parzygnat–Buscemi [6])). *Let* C *be the category of finite-dimensional* C^**-algebras with state-preserving CP maps as morphisms. A* retrodiction functor *is a contravariant functor* $R : \mathsf{C} \to \mathsf{C}^{\mathrm{op}}$ *satisfying:*

(i) Preservation: *R sends $(\mathcal{N}, \sigma)$ to a CPTP map.*

(ii) Bayesian consistency: *For composable channels $\mathcal{N}_1, \mathcal{N}_2$ and prior σ,*

$$R(\mathcal{N}_2 \circ \mathcal{N}_1,\, \sigma) = R(\mathcal{N}_1,\, \sigma) \circ R(\mathcal{N}_2,\, \mathcal{N}_1(\sigma)). \tag{3}$$

(iii) Normalization: $R(\mathrm{id}, \sigma) = \mathrm{id}$.

(iv) Classical limit: *On commutative subalgebras, R reduces to classical Bayes' rule.*

Then $R(\mathcal{N}, \sigma) = \mathcal{R}_{\sigma,\mathcal{N}}$ is the unique such functor.

Proof sketch.—Axioms (ii) and (iv) force the $\sigma^{1/2}(\cdot)\sigma^{1/2}$ sandwich structure. For commutative algebras, R must be the Bayesian inverse $R(N, p)(q)_i = p_i N_{j|i}/(\sum_k p_k N_{j|k})$, which in matrix form is $p^{1/2} N^T (Np)^{-1/2}(\cdot)(Np)^{-1/2} N p^{1/2}$. By the functoriality axiom (3) and the Stinespring dilation theorem, this extends uniquely to the noncommutative setting as Eq. (2). The full proof is in Ref. [6]; see also [9] for the Bayesian framework and Supplemental Material [34] for a self-contained derivation. □

Quantum eraser as Petz recovery.—In the quantum eraser, a signal-idler pair is prepared in the entangled state $|\psi\rangle_{SE} = (|A, d_A\rangle + |B, d_B\rangle)/\sqrt{2}$, where $|A\rangle, |B\rangle$ are path states and $|d_A\rangle, |d_B\rangle$ are orthogonal detector states. The which-path channel is $\mathcal{N} = \mathrm{Tr}_E$, giving the fully decohered signal state $\rho_S = \frac{1}{2}(|A\rangle\langle A| + |B\rangle\langle B|)$. Applying the Petz recovery (2) with $\sigma = |\psi\rangle\langle\psi|_{SE}$ and $\mathcal{N} = \mathrm{Tr}_E$:

$$\mathcal{R}_{\sigma,\mathrm{Tr}_E}(\rho_S) = |\psi\rangle\langle\psi|_{SE}, \tag{4}$$

which is exact because Tr_E saturates the DPI for pure bipartite states [34]. The "erasure measurement" on the idler (projecting onto $(|d_A\rangle \pm |d_B\rangle)/\sqrt{2}$) is the physical implementation of this Petz recovery. Strictly, the idler measurement is a *quantum instrument* (a collection of CP maps summing to a CPTP map); conditioned on outcome $\pm$, the effective map on the signal is CP but not trace-preserving. The Petz map $\mathcal{R}_{\sigma,\mathrm{Tr}_E}$ itself is CPTP and recovers the full entangled state (4); the subsequent projection onto $|\pm\rangle$ selects the subensemble where the coherent superposition $|A\rangle \pm |B\rangle$ is restored. The functoriality axiom (3) corresponds to the sequential structure of the eraser: composing the which-path marking with the idler measurement decomposes as retrodiction from the idler, then from the path.

Connection to TSVF.—The Aharonov–Bergmann–Lebowitz two-state vector formalism [14] assigns both a forward-evolving state $|\psi\rangle$ and a backward-evolving state $\langle\phi|$. Theorem 1 suggests a canonical choice for the backward state: the Petz-retrodicted state.

FROM RETRODICTION TO ERROR CORRECTION

The uniqueness of the Petz map as retrodiction functor connects naturally to its role as a QEC decoder. The following collects known results; our contribution is the interpretive link.

Theorem 2 (Near-Optimal QEC via Petz Recovery (Barnum–Knill [5], Junge et al. [11])). *Let $\mathcal{N}$ be a noise channel acting on a $[\![n, k, d]\!]$ stabilizer code $\mathcal{C}$ with code projector P and maximally mixed code state $\sigma_{\mathcal{C}} = P/\dim\mathcal{C}$. Let $\epsilon_{\mathrm{KL}} = \max_{a,b} \left\| P E_a^\dagger E_b P - c_{ab} P \right\|$ measure deviation from the Knill–Laflamme conditions [10], where $\{E_a\}$ are Kraus operators of $\mathcal{N}$. Then:*

(a) *If $\epsilon_{\mathrm{KL}} = 0$ (exact QEC), $\mathcal{R}_{\sigma_{\mathcal{C}},\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on $\mathcal{C}$.*

(b) *In general, the universal recovery theorem of Junge et al. [11] (Corollary 5.1 therein) guarantees the existence of a* rotated *Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ such that:*

$$F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma_{\mathcal{C}},\mathcal{N}}(\mathcal{N}(\rho))\big)^2 \geq \exp\big(-\Delta D\big), \tag{5}$$

where $\Delta D = D(\rho\|\sigma_{\mathcal{C}}) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ is the relative entropy decrease. The standard Petz map satisfies the weaker Barnum–Knill bound [5]: $F_e \geq \frac{1}{2} F_{e,\mathrm{opt}}^2$.

Proof.—(a) When the Knill–Laflamme conditions hold exactly, the DPI is saturated for all code states: $D(\rho\|\sigma_{\mathcal{C}}) = D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ for all ρ supported on $\mathcal{C}$. By Petz's sufficiency theorem [4], the Petz recovery is exact. (b) The rotated Petz map is $\tilde{\mathcal{R}}(\cdot) = \int dt\; p(t)\, \sigma^{it} \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\sigma)^{-it}(\cdot)\mathcal{N}(\sigma)^{it})\sigma^{-it}$, where $p(t) = (\pi/2)(\cosh(\pi t) + 1)^{-1}$; the bound follows from Ref. [11]. □

In this light, QEC decoding is retrodiction: the decoder retrodicts the original ρ from the corrupted $\mathcal{N}(\rho)$ using the code state as prior. The Petz map is the unique Bayesian-consistent choice (Theorem 1), and its near-optimality [5] follows from this uniqueness rather than from case-specific linear algebra.

THE THERMODYNAMIC STRUCTURE

When the reference state is thermal, known results combine into a chain linking recovery fidelity to entropy production. We state this chain to establish the connection to the temporal asymmetry parameter τ.

Theorem 3 (Master Inequality Chain (assembling [11–13, 18])). *For any quantum channel $\mathcal{N}$, reference state σ, and state ρ:*

$$\boxed{-\log F^2 \;\le\; I(A;E|B)_\omega \;\le\; \Sigma \;\le\; \Delta D} \tag{6}$$

where $F = F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$ is the recovery fidelity; $I(A;E|B)_\omega$ is the conditional mutual information of the Stinespring complementary output; Σ is the entropy production (Kolchinsky–Wolpert decomposition [13]); and $\Delta D = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ is the relative entropy decrease. When $\sigma = e^{-\beta H}/Z$ is a Gibbs state:

$$\Delta D = \beta\, W_{\rm diss} = \Delta S_{\rm irr}/k_{\rm B}, \tag{7}$$

yielding a thermodynamic bound on recovery: $F^2 \ge e^{-\Delta S_{\rm irr}/k_{\rm B}}$.

Proof.—Each step uses a known result; our contribution is the assembly into a single chain with the τ interpretation. Here F denotes the fidelity with the rotated Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ of Ref. [11].

Step 1 (Fidelity bound): By the universal recovery theorem [11], $F^2 \ge \exp(-\Delta D)$, so $-\log F^2 \le \Delta D$.

Step 2 (CMI bound): By Fawzi–Renner [12], $I(A;E|B) \ge -\log F^2$ via the strengthened data processing inequality applied to the Stinespring dilation, giving the leftmost bound.

Step 3 (Entropy production): For a thermodynamic process with Gibbs reference state $\sigma = e^{-\beta H}/Z$, the entropy production satisfies $\Sigma \ge I(A;E|B)$ because the Landauer component $\Sigma_{\rm Landauer} = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ equals $\Delta D \ge I(A;E|B)$ (by Steps 1–2), and $\Sigma \ge \Sigma_{\rm Landauer}$ by the Kolchinsky–Wolpert decomposition [13].

Step 4 (Second law): The relative entropy decrease $\Delta D \ge \Sigma$ by Reeb–Wolf [18]: the free-energy dissipation exceeds the entropy production by system-bath correlations and bath displacement.

The tightness conditions are physically transparent: $F^2 = 1$ iff exact QEC (Knill–Laflamme satisfied); $I(A;E|B) = 0$ iff Markov condition (no information leaks to environment); $\Sigma = 0$ iff thermodynamically reversible process. See Supplemental Material [34] for the detailed derivation including the Gibbs specialization (7). □

Quantum Crooks connection.—When $\sigma = \sigma_\beta = e^{-\beta H}/Z$, the Petz map becomes $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ with the structural factor $\sigma_\beta^{1/2} = e^{-\beta H/2}/\sqrt{Z}$ —exactly the "half Boltzmann weight" appearing in the KMS condition of modular theory. Kwon and Kim [7] proved that this $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ is precisely the reverse channel in the quantum Crooks fluctuation theorem:

$$\frac{p_F(w)}{p_R(-w)} = e^{\beta(w-\Delta F)}, \tag{8}$$

where $p_F(w)$ and $p_R(-w)$ are forward and Petz-reversed work distributions. The detailed balance condition $\mathcal{R}_{\sigma,\mathcal{N}} = \mathcal{N}$ characterizes thermodynamically reversible processes—those that are their own Petz reversal.

Temporal asymmetry parameter.—Define

$$\tau \;=\; 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big). \tag{9}$$

This parameter quantifies the degree to which the channel $\mathcal{N}$ distinguishes past from future when σ is the reference state. A remark on the two Petz maps is in order. The retrodiction uniqueness theorem [6] singles out the *standard* Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ as the unique retrodiction functor; the quantum Crooks theorem likewise uses $\mathcal{R}_{\sigma,\mathcal{N}}$ as the physical reverse channel. The bound (5) and the definition of τ require the *rotated* map $\tilde{\mathcal{R}}$, which averages $\mathcal{R}$ over modular automorphisms $\sigma^{it}(\cdot)\sigma^{-it}$. For a thermal reference $\sigma = e^{-\beta H}/Z$, the rotation $\sigma^{it} = e^{-it\beta H}/Z^{it}$ is simply unitary time evolution generated by H, so $\tilde{\mathcal{R}}$ smooths the standard Petz map over the system's own Hamiltonian dynamics. This thermal averaging handles quantum noncommutativity $[\rho,\sigma] \ne 0$ and ensures that the fidelity bound $F^2 \ge \exp(-\Delta D)$ holds universally. When $[\rho,\sigma] = 0$ (classical limit) or $\Delta D = 0$ (exact recovery), $\tilde{\mathcal{R}}$ reduces to $\mathcal{R}$ and the distinction vanishes. Three properties follow from the master chain (6):

(i) *Closed systems:* If $\mathcal{N}$ is unitary, $\mathcal{R}_{\sigma,\mathcal{U}} = \mathcal{U}^\dagger$ and $F = 1$, so $\tau = 0$. There is no temporal asymmetry.

(ii) *Bound by entropy production:* From $F^2 \ge \exp(-\Sigma)$,

$$\tau \;\le\; 1 - \exp(-\Sigma/2). \tag{10}$$

When $\Sigma = 0$ (zero entropy production), $\tau = 0$ is enforced.

(iii) *Equivalence chain:* For faithful σ (see [34], Sec. S11):

$$\tau = 0 \iff \Sigma = 0 \iff I(A;E|B) = 0 \iff \text{QMC}, \tag{11}$$

where QMC denotes the quantum Markov chain condition [12]. Each individual equivalence is established [4, 12, 29]; the full chain (11) states them as a single result, formalizing the physical content: the arrow of time ($\tau > 0$) is equivalent to environmental coupling ($I(A;E|B) > 0$), entropy production ($\Sigma > 0$), and imperfect recoverability ($F < 1$).

NEW RESULT I: BOUND SATURATION THEOREM

We identify sufficient conditions under which the Junge et al. bound $F^2 \ge \exp(-\Delta D)$ is exactly saturated.

Theorem 4 (Saturation of the Recovery Bound)**.** *Let $\mathcal{N}$ be a CPTP map, σ a full-rank reference state, and $\rho = |\psi\rangle\langle\psi|$ a pure state. If the following conditions hold:*

(i) $\mathcal{N}(\rho) = \rho$ (fixed point);

(ii) $[\rho, \sigma] = 0$ (input commutes with reference);

(iii) $[\rho, \mathcal{N}(\sigma)] = 0$ (input commutes with channel output of reference);

then the Junge et al. bound is exactly saturated:

$$F^2\big(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big) = \exp(-\Delta D) = \frac{\langle\psi|\sigma|\psi\rangle}{\langle\psi|\mathcal{N}(\sigma)|\psi\rangle}. \quad (12)$$

Proof sketch.—Work in a basis where $|\psi\rangle = |0\rangle$. Condition (ii) gives $\sigma = s_0|0\rangle\langle 0| \oplus \sigma_\perp$; condition (i) forces $E_k|0\rangle = \lambda_k|0\rangle$ with $\sum_k |\lambda_k|^2 = 1$. Then $\Delta D = \ln(\omega_0/s_0)$ where $\omega_0 = \langle 0|\mathcal{N}(\sigma)|0\rangle$, and direct computation of the Petz map gives $F^2 = s_0/\omega_0 = \exp(-\Delta D)$. See [34], Sec. S16–S17, for the full derivation and numerical verification across six channel families. □

Bayesian interpretation.—The saturation value $F^2 = s_0/\omega_0$ is the ratio of prior to posterior probability: $P(\psi|\text{prior})/P(\psi|\text{posterior})$. Saturation occurs precisely when quantum recovery reduces to classical Bayesian retrodiction on the one-dimensional subspace spanned by $|\psi\rangle$.

Examples.—(a) Amplitude damping $\mathcal{N}_\gamma^{\text{AD}}$ with $\rho = |0\rangle\langle 0|$ (ground state) and $\sigma = I/2$: $F^2 = 1/(1+\gamma) = \exp(-\ln(1+\gamma))$, saturated for all $\gamma \in [0, 1]$. (b) Pure loss bosonic channel (transmissivity η) with vacuum input: saturated for all η. (c) The complement of the amplitude damping channel (which is *anti-degradable*) also saturates—degradability is irrelevant. (d) Unital channels (dephasing, depolarizing) are *strictly non-saturated* in the interior [34].

Remark.—Conditions (i)–(iii) are sufficient but not necessary. The completely depolarizing channel $\mathcal{N}(\rho) = I/d$ saturates for all pure inputs ($F^2 = 1/d = \exp(-\ln d)$) despite condition (i) failing. A complete characterization of necessity remains open.

Experimental relevance.—Amplitude damping is the dominant noise in superconducting qubits (T_1 decay) and ion traps. Petz recovery has been implemented on both platforms [32, 33]. The prediction $F^2 = 1/(1+\gamma)$ for ground-state input is directly testable.

NEW RESULT II: COMPOSITIONAL SUB-ADDITIVITY OF τ

The Petz map's functoriality (3) implies a geometric structure on τ that other recovery maps lack.

Theorem 5 (Composition Sub-additivity)**.** *For sequential channels $\mathcal{N}_1, \mathcal{N}_2$ and reference state σ, the Petz recovery τ satisfies:*

$$\sqrt{\tau_{12}} \;\le\; \sqrt{\tau_1} + \sqrt{\tau_2^{\text{eff}}}, \quad (13)$$

where $\tau_{12} = 1 - F(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}_2\circ\mathcal{N}_1} \circ (\mathcal{N}_2\circ\mathcal{N}_1)(\rho))$, $\tau_1 = 1 - F(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}_1}\circ\mathcal{N}_1(\rho))$, and $\tau_2^{\text{eff}} = 1 - F(\mathcal{N}_1(\rho),\, \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2}\circ \mathcal{N}_2(\mathcal{N}_1(\rho)))$ is the second-stage cost evaluated at the evolved *state $\mathcal{N}_1(\rho)$ with* updated *reference $\mathcal{N}_1(\sigma)$.*

Proof.—Three ingredients. (1) *Functoriality* (3): $\mathcal{R}_{\sigma,\mathcal{N}_2\circ\mathcal{N}_1} = \mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2}$. (2) *Bures triangle inequality*: for any three states ρ, ρ_1, ρ_{12}, $\sqrt{1 - F(\rho, \rho_{12})} \le \sqrt{1 - F(\rho, \rho_1)} + \sqrt{1 - F(\rho_1, \rho_{12})}$. (3) *Data-processing inequality*: $\mathcal{R}_{\sigma,\mathcal{N}_1}$ is CPTP, so $F(\rho_1, \rho_{12}) = F(\mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2} \circ \mathcal{N}_2 \circ \mathcal{N}_1(\rho),\, \mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{N}_1(\rho)) \ge F(\mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2} \circ \mathcal{N}_2(\mathcal{N}_1(\rho)),\, \mathcal{N}_1(\rho))$. Combining gives (13). See [34], Sec. S18. □

Remark.—The reference for the second stage *must* be the Bayesian-updated $\mathcal{N}_1(\sigma)$, not the original σ. With the original reference, the inequality fails numerically. This is physically meaningful: irreversibility decomposes additively across sequential processes only when each stage's cost is measured using the updated prior—the hallmark of Bayesian inference.

Uniqueness to Petz.—The rotated Petz map $\tilde{\mathcal{R}}$ of Ref. [11], which gives tighter fidelity bounds, does *not* satisfy Eq. (13): numerical tests find violations for non-unital channel pairs (see [34], Sec. S18). This provides a new structural distinction between the two recovery maps beyond what the categorical axioms [6] alone imply.

OBSERVATIONS FROM THE τ FRAMEWORK

Two further observations follow from interpreting existing results through τ.

Observation 1: Post-selection as thermodynamic filtering.—For a stabilizer code under noise $\mathcal{N}$, each syndrome s has a data-processing gap $\Delta D(s)$. Since $\Delta D(s)$ depends only on the noise channel and code state, the per-distance scaling of rejection rates should be decoder-independent. Numerical simulations at $d \in \{3, 5, 7, 9, 11\}$ with three decoders yield a coefficient of variation CV = 0.091 across decoders, consistent with this expectation [34]. This is also consistent with the Ising–QEC mapping [21], statistical-mechanical thresholds [20], and scalable post-selection results [19, 22].

Observation 2: Decoder hierarchy as retrodiction approximation.—Since the Petz map is the unique Bayesian-consistent retrodiction (Theorem 1), all good decoders must approximate it. The known decoder hierarchy [23]—ML > neural network [24] > MWPM > Union-Find—is consistent with being ordered by retrodiction fidelity. A direct test would require computing

the channel divergence $D(\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}\|\mathcal{D})$ for each decoder $\mathcal{D}$, which we leave to future work.

THE QUANTUM ERASER REVISITED

The quantum eraser is a pure bipartite system: $\mathcal{N} = \mathrm{Tr}_E$, $\sigma = |\psi\rangle\langle\psi|_{SE}$. As computed in [34] (Sec. S1), the Petz map achieves $F = 1$, hence $\tau = 0$. This is a direct consequence of the equivalence chain (11): for a pure bipartite state, $I(A; E|B) = 0$ (no environment beyond the idler) and $\Sigma = 0$ (no dissipation). The quantum eraser "works" because the signal-idler pair forms a closed system with $\tau = 0$.

Superposition necessity.—By contrapositive: if a system is in a definite path state, the Petz recovery produces no coherence (see [34], Sec. S5). Recovery of interference implies prior superposition.

Wheeler's delayed choice and τ.—In Wheeler's delayed-choice experiment [15], the signal-idler pair forms a time-symmetric ($\tau = 0$) system where forward and reverse processes are statistically indistinguishable. The *order* of measurements is operationally irrelevant: measuring the signal first or the idler first yields identical statistics. In the τ language, the delayed-choice aspect requires no retrocausal explanation: the closed system has $\tau = 0$, so the temporal order of measurements is irrelevant—a conclusion consistent with the standard decoherence perspective [27]. The "need to wait" for the idler measurement arises only in the open-system description, where one traces out the idler and observes $\tau > 0$ (decoherence) [27].

DISCUSSION

The interpretive framework presented here rests on known mathematical results—the Petz map's uniqueness as retrodiction functor [6], the universal recovery bound [11], and the connection to entropy production [7, 13, 26]. Our contributions are: (1) the definition of $\tau = 1-F$ as a continuous measure of the arrow of time; (2) sufficient conditions for exact saturation of the recovery bound, with a Bayesian interpretation (Theorem 4); and (3) a composition sub-additivity theorem showing that $\sqrt{\tau}$ forms a metric-like quantity unique to the Petz map (Theorem 5).

The τ parameter provides a unified language for phenomena across quantum information:

Domain	**Key result**	τ
Quantum eraser	$F = 1$ (exact)	0
Closed unitary	Petz = inverse	0
QEC [11]	$F^2 \geq e^{-\Delta D}$	> 0
Thermodynamics	Eq. (10)	$\leq 1 - e^{-\Sigma/2}$
Saturation (new)	Thm. 4	$= 1 - \sqrt{s_0/\omega_0}$
Composition (new)	Thm. 5	$\sqrt{\tau_{12}} \leq \sqrt{\tau_1} + \sqrt{\tau_2^{\mathrm{eff}}}$

The physical picture connects to the decoherence program [27, 28]: environment-induced decoherence produces entropy ($\Sigma > 0$), implying $\tau > 0$. The arrow of time, in this interpretation, is a consequence of system-environment coupling quantified by Petz recovery failure.

Limitations.—The sufficient conditions in Theorem 4 are not necessary: the completely depolarizing channel saturates without satisfying condition (i). A full characterization of necessity remains open. The composition inequality (Theorem 5) holds for the standard Petz map; its generalization to approximate or smoothed recovery maps is not established. The QEC observations (Sec.) are numerically supported but lack rigorous proofs.

Open questions.—(i) What are the necessary and sufficient conditions for saturation of the Junge et al. bound? (ii) Does the composition sub-additivity extend to continuous-time Lindbladian evolution? (iii) For non-Markovian processes, Σ can be temporarily negative [30], implying transient decreases in τ. Can the process tensor framework [16] extend the equivalence chain (11)? (iv) Since entanglement wedge reconstruction uses the Petz map [8], does τ have a holographic interpretation?

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[34] See Supplemental Material for detailed derivations, the explicit quantum eraser calculation, the saturation theorem proof (Sec. S17), the composition sub-additivity proof (Sec. S18), post-selection analysis, the equivalence chain proof, and caveats.